

For a basket of stock the market impact cost is:

$$MI(\alpha_i) = \sum_{i=1}^m b_1 \cdot I_i^* \alpha_i \quad (9.13)$$

Timing Risk Expression

The timing risk for a portfolio cannot be expressed as a continuous function in terms of the trading rate parameter because at some time the residual shares would fall below zero. But we can overcome this problem by approximating the residual with the following continuous exponential function:

$$r_{ij} = X_i \cdot e^{-j\gamma_i} \quad (9.14)$$

where,

$$\gamma_i = 2.74 \cdot T_i^{-1.22} + 0.01 \quad (9.15)$$

This representation of residuals results in approximately the same risk that is computed using the trade schedule strategy.

The trade rate optimization problem is formulated as follows:

$$\text{Min} \sum_{i=1}^m b_1 \cdot I_i^* \alpha_i + \lambda \sqrt{r^t Cr} \quad (9.16)$$

Subject to:

$r_{ij} = X_i \cdot e^{-j\gamma_i}$	for all i, j
$\gamma_i = 2.74 \cdot T_i^{-1.22} + 0.01$	for all i
$T_i = \frac{X_i}{ADV_i} \cdot \frac{1}{\alpha} \cdot n$	for all i
$LB_i \leq \alpha_i \leq UB_i$	for all i, j

The LB needs to be set at a value that will ensure the order will be completed by the investor's specified end time.

Representation of the trade schedule in terms of trading rate provides many benefits. There is only one parameter per stock. The market impact cost and timing risk expressions are greatly simplified. Completion of the order and the shrinking portfolio constraint are guaranteed. And since the gradient and Hessian are easily computable, it provides efficiency and speed for the non-linear optimization.

A limitation of the trade rate formulation is that it does not provide as much freedom to take complete advantage of risk reduction opportunities